Bridging the Sim-to-Real Gap for Systems with Guarantees

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Problem

- Co-Design allows to concurrently optimize system morphology and control to satisfy specific guarantees, such as stability
- Problem: Co-Design is done almost entirely **in simulation**
  - Simulation models might not capture the system's complexity
  - Changing morphology means that different system dynamics may dominate after optimization
  - Badly identified dynamics can corrupt the results

→ In practice, theoretical guarantees might not transfer to the physical system

Closing the Sim-to-Real Gap

- How can **real-world data** be used **in the loop** of the codesign process?
- How can the **amount of experiments** be kept as small as possible?
- How can the **difference** between simulation and real be **interpreted**?

Parametrizing Robustness [1]

We consider a non-linear torque-controlled inverted pendulum with state $x = [\theta, \dot{\theta}]^T \in \mathcal{X} \subseteq \mathbb{R}^2$, input $u \in \mathcal{U} \subseteq \mathbb{R}$ and equilibrium $x^* = \left(\pi, 0\right)^T$. Linearizing the system and using an LQR controller with gains Q and R, the Lyapunov function $V(x) = x^T P x$ guarantees stability of the linearized, unconstrained system.

A guarantee for the full non-linear closed-loop system is however only possible in a region around the setpoint x^*, called the *region of attraction (RoA)*. In general, the RoA cannot be calculated explicitly, and is in practice often approximated as the sublevel set of a Lyapunov function

$$\mathcal{R}_a := \{x \in \mathcal{X} | x \to x^* \text{ for } t \to \infty\} \approx \{x | V(x) < \rho\}, \tag{1}$$

where $\rho > 0$. This is a valid RoA estimate as long as $V(x) > 0$ and $\dot{V}(x) < 0$ for all $x \in \mathcal{R}_a \setminus \{0\}$, and the best approximation is found by calculating the maximum ρ satisfying these conditions.

Since the RoA gives a strong theoretical guarantee for stability, its volume can be used as a proxy for robustness and safety: A larger volume means higher robustness to disturbances and a larger safe region of operation.

Co-Design Formulation

Parameters:

- Morphology: Mass m of the system
- Control: LQR-controller gains Q and R

Parametrization to (a) avoid over-parametrization and (b) ensure well-definedness of Q and R:

$$R = \text{const.}$$
$$Q = \begin{pmatrix} q_1 & 0 \\ 0 & q_2 \end{pmatrix} = \begin{pmatrix} q_1^{max} e^{-\bar{q}_1} & 0 \\ 0 & q_2^{max} e^{-\bar{q}_2} \end{pmatrix} \tag{2}$$

Objective: Maximize the *volume* of the approximated RoA as a proxy for stability and robustness: With $V(x) = (x - x^*)^T P (x - x^*)$ and the eigendecomposition $\frac{1}{\rho} P = W^T \Lambda W$, calculate

$$\text{vol}(\mathcal{R}_a) = |\det(\sqrt{\Lambda} W)|^{-1} \frac{\pi}{2}. \tag{3}$$

Optimization Problem: The full optimization problem becomes

$$\max_{m, \bar{q}_1, \bar{q}_2} \text{vol}(\mathcal{R}_a) \tag{4a}$$
$$\text{subject to } \; m \in [m_{min}, m_{max}] \tag{4b}$$
$$\bar{q}_i \in \left[0, -\log \left(\frac{q_1^{min}}{q_1^{max}} \right) \right] \tag{4c}$$

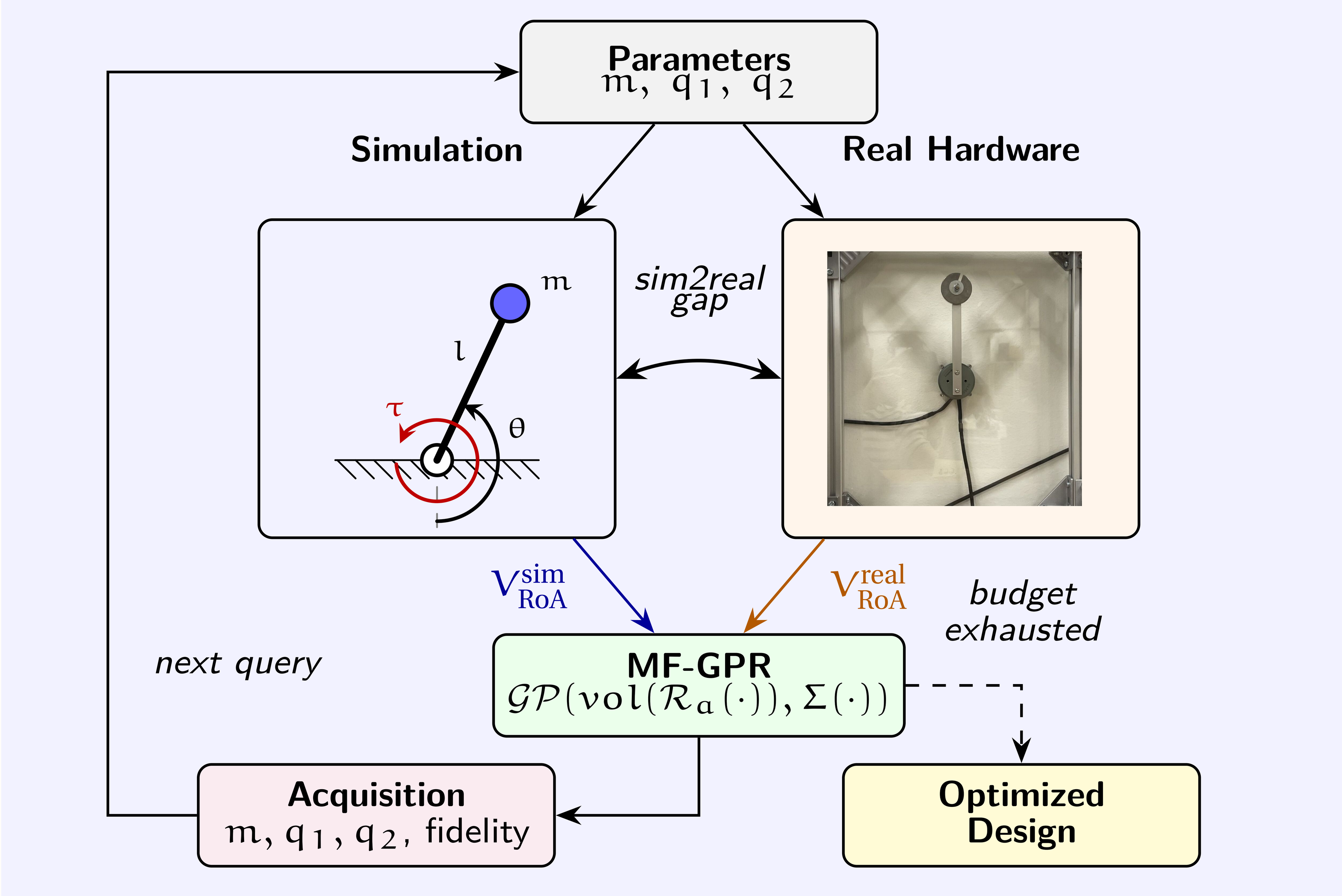
where $\text{vol}(\mathcal{R}_a)$ is evaluated with respect to the system dynamics **??** at the chosen mass m and an LQR controller parametrized by Eq. (2).

Optimization Algorithm:

- With a *gradient-free, sampling-based* optimization algorithm, the Co-Design Formulation can run at *any fidelity level* (i.e. real system, different system models).
- We run it instead at *multiple fidelities concurrently*.

To this end, we adopt a Multi-Fidelity Gaussian Process Regression (MF-GPR) as described in [2]. The idea is to find a surrogate model for the map $f_i(m, q_1, q_2) \to \text{vol}(\mathcal{R}_a)$ at each fidelity i that consists of multiple latent Gaussian Processes and a structure that describes the relationship between them. The step from value-estimation to optimization happens through a so-called acquisition function that selects the next test points that maximizes the information about the maximum of $f_i(m, q_1, q_2)$.

Framework Overview



Results

- Validation: torque-controlled simple pendulum ($m \in [70\,\text{g}, 150\,\text{g}]$, $l = 0.05\,\text{m}$, $u \in \pm 0.04\,\text{N m}$)
- 2 fidelities: "real" and "simulation"
- Simulation uses inaccurate model (disregard friction, damping, high noise in velocity measurements)
- Initialized with 25 simulations and 3 experiments (cost 1 vs. 5) and runs until a total cost budget of 150 is exhausted
- Stops after 52 iterations (199 sims, 28 experiments / ca. 52 min of experiment data)
- Results ($m = 70\,\text{g}$, $q_1 = 7.3 \times 10^{-4}$, $q_2 = 1.3 \times 10^{-6}$) are reasonable
- Validation against entirely simulation-based method (using CMA-ES [1]): $\text{vol}(\mathcal{R}_a^{MF-GPR}) = 47.43$ (ours) vs $\text{vol}(\mathcal{R}_a^{CMA-ES}) = 28.22$ (theirs)

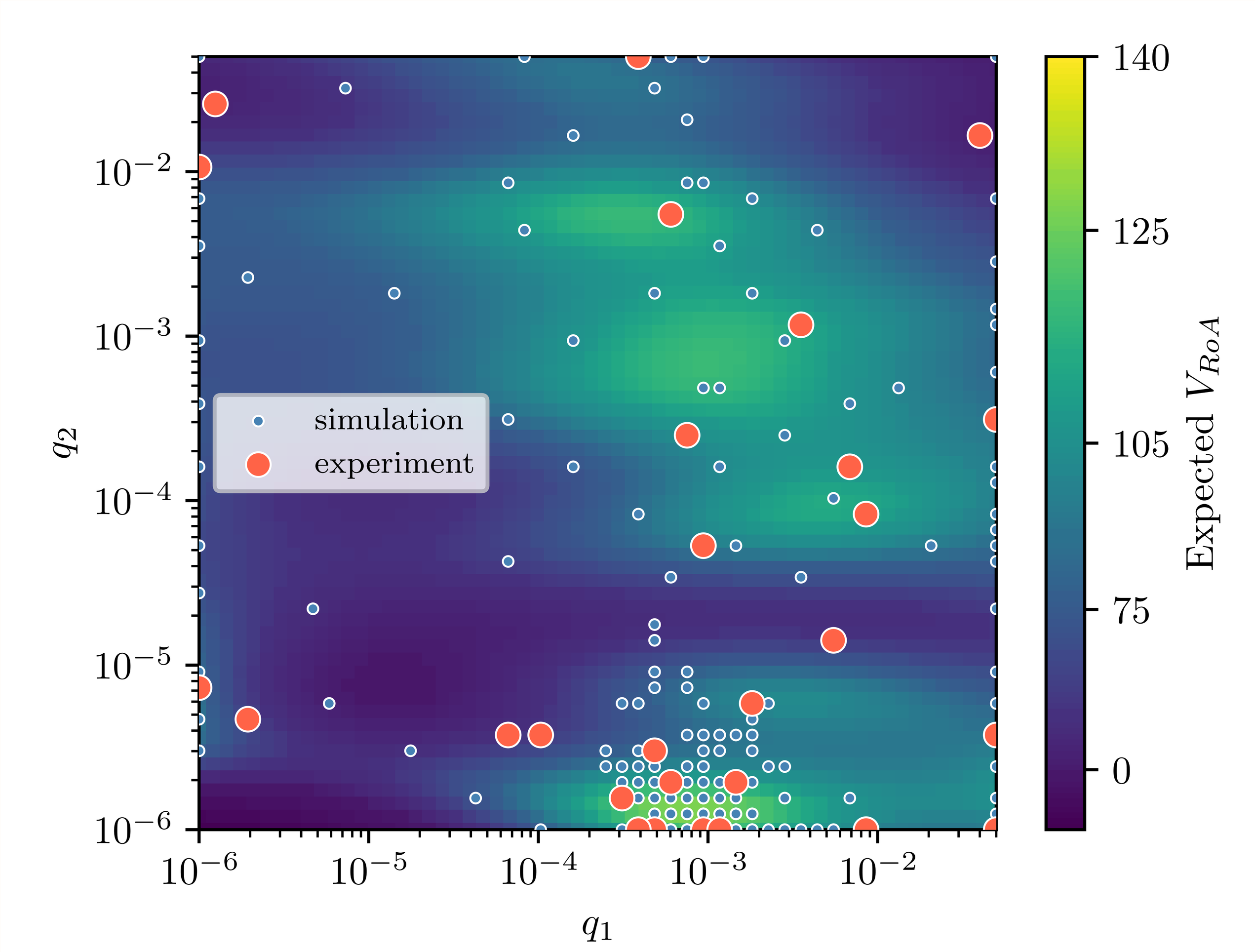


Figure 1: Estimated Volume of the RoA at "experiment" fidelity for q_1, q_2 with $m = 70\,\text{g}$.

Method	q_1	q_2
MF-GPR	7.3×10^{-4}	1.3×10^{-6}
CMA-ES	2.7×10^{-3}	3.2×10^{-4}
Baseline	1×10^{-4}	1×10^{-6}

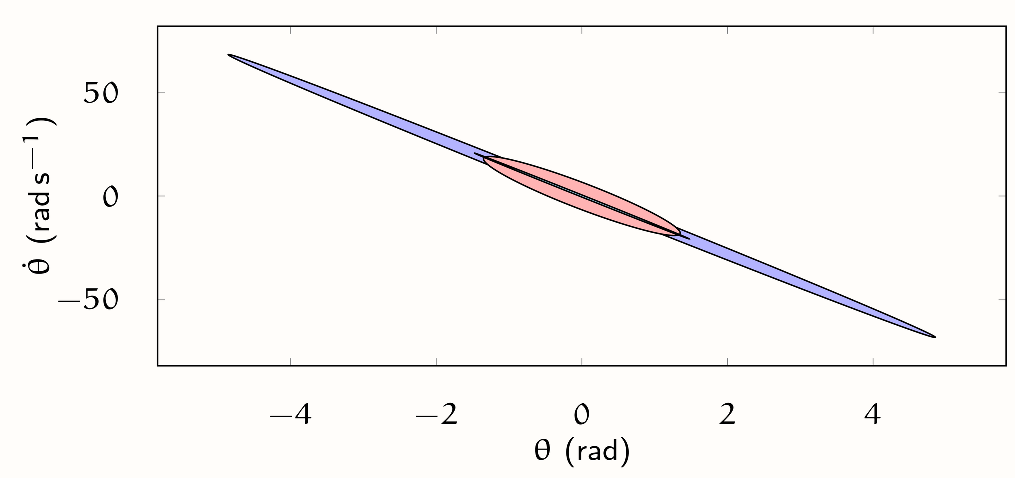
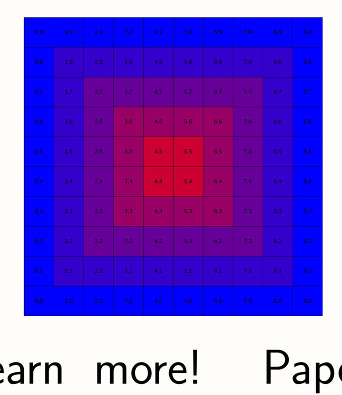


Figure 2: Resulting $\mathcal{R}_a$ as measured on the hardware.

[1] Lasse Jenning Maywald et al. "Co-optimization of Acrobot Design and Controller for Increased Certifiable Stability". In: *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2022, pp. 2636–2641.

[2] Shion Takeno et al. "Multi-fidelity Bayesian optimization with max-value entropy search and its parallelization". In: *International Conference on Machine Learning*. PMLR. 2020, pp. 9334–9345.



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